



$$\beta = \frac{R_2}{R_1 + R_2}$$

$$\frac{V_{out}(s)}{V_{in}(s)} = \frac{A(s)}{1 + \beta A(s)}$$

$$= \frac{A_0 / (1 + s/\omega_0)}{1 + \beta \frac{A_0}{(1 + s/\omega_0)}}$$

$$= \frac{A_0}{1 + \frac{s}{\omega_0} + \beta A_0}$$

$$= \frac{A_0}{(1 + \beta A_0) \left[1 + \frac{s}{(1 + \beta A_0)\omega_0} \right]}$$

$$= \frac{A_0 / (1 + \beta A_0) \Rightarrow A_{dc, closed}}{1 + \frac{s}{(1 + \beta A_0)\omega_0} \Rightarrow s/\omega_{u, closed}} = \frac{A_{dc, closed}}{1 + s\tau}$$

$$\tau = \frac{1}{(1 + \beta A_0)\omega_0} \approx \frac{1}{\beta A_0 \omega_0} \approx \frac{1}{\beta \omega_u}$$

$$A_{dc, closed} = \frac{A_0}{1 + \beta A_0} \approx \frac{1}{\beta} = \frac{R_1 + R_2}{R_2} = 1 + \frac{R_1}{R_2} \approx 10$$

$$\frac{V_{out}(s)}{V_{in}(s)} = \frac{A_{dc, closed}}{1 + s\tau}$$

$$V_{out}(s) = \frac{A_{dc, closed}}{1 + s\tau} \cdot \frac{a}{s}$$

$$= a A_{dc, closed} \left[\frac{1}{s(1 + s\tau)} \right]$$

$$= a A_{dc, closed} \left[\frac{1}{s} + \frac{-\tau}{1 + s\tau} \right]$$

$$= a A_{dc, closed} \left[\frac{1}{s} - \frac{1}{\frac{1}{\tau} + s} \right]$$

$$V_{out}(t) = a A_{dc, closed} \left[1 - e^{-t/\tau} \right] u(t)$$

$$\frac{1}{s(1 + s\tau)} = \frac{A}{s} + \frac{B}{1 + s\tau}$$

$$A(1 + s\tau) + Bs = 1$$

$$A = 1$$

$$1 + s\tau = 0$$

$$s = -1/\tau$$

$$B = -\tau$$

$$a A_{dc, closed} \times 0.99 = a A_{dc, closed} \left[1 - e^{-\frac{t}{\tau} \times 0.99} \right]$$

$$e^{-t/\tau} = 0.01$$

$$\frac{t}{\tau} = 4.605$$

$$t = 4.605 \tau$$

$$t = \frac{4.605}{\beta \omega_u}$$

$$t = \frac{4.605}{0.1 \omega_u}$$

$$t < 5\text{ns}$$

$$\frac{4.605}{0.1 \omega_u} < 5 \times 10^{-9}$$

$$\frac{4.605}{0.1 \times 5 \times 10^{-9}} < \omega_u$$

$$\omega_u > 9.21 \times 10^9 \text{ rad/sec}$$

$$2\pi f_u > 9.21 \times 10^9$$

$$f_u > \frac{9.21 \times 10^9}{2\pi} = 1.47 \text{ GHz}$$